



Facebook Bug made posts public

Facebook recently revealed that it encountered a glitch in May that made up to 14 million users' posts set to public. <https://dgit.in/jul18-26-1>



Headphone jack module

Essential lets users to add a headphone jack module to their Essential Phone. <https://dgit.in/jul18-26-2>

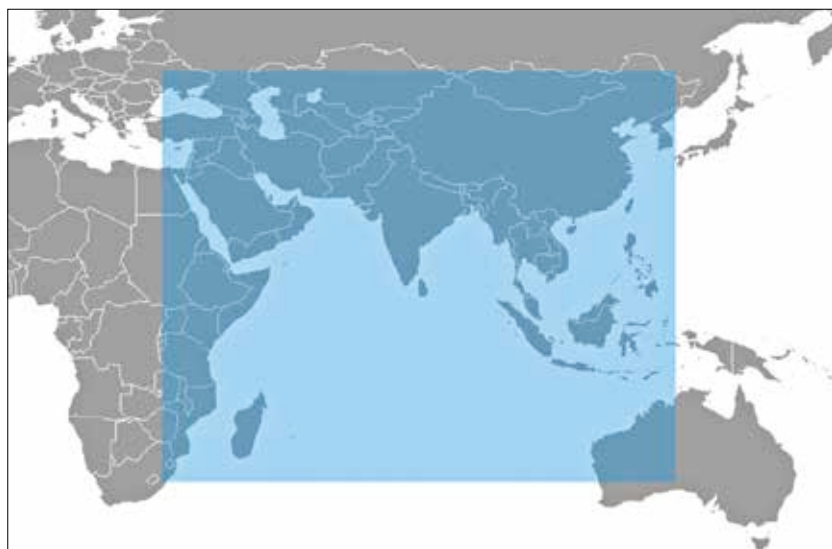
during one of the integrated electrical checks. It was conducted successfully on March 28th.

HICCUPS ON THE WAY

The next few launches of the IRNSS project did not see any problems save for operational delays. IRNSS 1E was launched in January 2016, the 1F in March 2016 and the final part of the constellation, the IRNSS 1G, was successfully launched on April 28 2016 aboard the PSLV C33. With this, the system was also renamed to NavIC, or Navigation with Indian Constellation, by the Prime Minister of India, Narendra Modi. At this point, the IRNSS was considered complete and would have been operational, giving India its

the two satellites available as backup. Unfortunately, all wasn't to go well for this launch.

On 31st August, 2017, during the final stage of the launch, the heat shield of the rocket failed to open up for the separation of the satellite. A heat shield is that component of the rocket that keeps its precious cargo safe from external heat generated due to friction in launch period. The heat shield is at the very top of the rocket, and is also known as the nose cone or the payload fairing. It is supposed to open up like a flower, and release the satellite into space once the conditions are suitable at the right time during a launch. This was the PSLV's first failure since its inaugural attempt in 1993.



The coverage area of IRNSS

own navigation system. However, the program ran into a couple of problems after that.

Over a time period of six months, three of the rubidium atomic clocks aboard the IRNSS 1A failed. Coincidentally, around the same time, 10 atomic clocks used by the Galileo navigation project by the European Space Agency also failed. In both cases, the clocks were supplied by Spectratime. While there was no particular conclusion on what was to blame for these consecutive failures, a replacement satellite launch for the IRNSS 1H was prepared for launch. This was one of

Additionally, in March 2018, the Comptroller General of India (CAG) flagged cost overruns and significant delays in the overall IRNSS project in a report tabled in Parliament. While the initial budget set by the Government of India was ₹1,420 crores (approx), the overall project expenditure had reached ₹2,446 crores (approx) by March 2017, of which ₹1,284 crores was spent under the program, and an additional expenditure of ₹1,162 crores was incurred on launch vehicles and maintenance of the satellites and the ground segment. For the 45 contracts awarded for the realisation of several

components of the program, the CAG report observed delays in execution ranging from three months to nearly nine years. ISRO also drew some flak for its refusal to develop the user segment (essentially the schematics for the NavIC chips) since they felt it was the responsibility of the user - although it was included as part of the project in their initial budget approval. Work on this was started in March 2017.

IT'S ALIVE!

On April 12, 2018, the replacement (of the replacement) for the IRNSS 1A, the IRNSS 1I was successfully launched. Interestingly, the 1A was never decommissioned and is still in use for messaging purposes. There are two IRNSS satellites kept ready on the ground at all times, to act as replacements in case any of the existing satellites fail. As of June 2018, ISRO had asked for Request of Proposal (RoP), for the implementation of NavIC from vendors. The scope of work is supposed to be the following:

- Development of the NavIC receiver hardware for two types based on SAC requirements.
- Development of the NavIC receiver hardware based on SAC requirements, which includes component selection, schematic capture, PCB layout, procurement of components, fabrication of PCB and mounting components and test the receiver.
- Design and fabrication of suitable and compact enclosure for Receiver. It should provide proper RF shielding and EMI/EMC performance.
- Preparation of FAT/ATP document, to carry out FAT in presence of ISRO officials.
- Production of 2 PROTO Units for each type followed by ordered number of units.

In the near future, NavIC may not just come to the aid of naval and aerial navigation, but also provide a replacement for the GPS chips that we use on our smartphones and in our cars.